# ENDOMETRIOSIS

**Definition :** Presence of functional endometrial tissues in abnormal sites.

***The term adenomyosis:*** endometriosis of uterine muscles

**Incidence :** 5-8% of women .

**Types :** 2 types of endometriosis

| Pelvic type | extra-pelvic type |
|---|---|
| 1- Ovaries: most common site | 1- Umblicus |
| 2- Uterus: myometruim. | 2- Other organs e.g. lung. |
| 3- Tubes. | 3- Laparotomies scar |
| 4-Uterine ligaments: uterosacral ligament. | 4- Episiotomy scar |
| 5- Pelvic peritoneum: e.g. Douglas pouch | 5- C.S. scar. |
| 6- Vagina and rectovaginal septum | 6- Laparoscopy scar. |
| 7- Vulva | |
| 8- Rectum | |
| 9- Bladder | |

## Pelvic Endometriosis

**Theories (Histogenesis) of endometriosis:**

***1- Tubal regurgitation theory of Sampson:***

Fragments of endometrium implants in the pelvis through retrograde menstruation.

***2- Meyer's serosal metaplasia theory:***

The peritoneal epithelium can undergo metaplasia into endometrial epithelium in response to some abnormal stimuli.

***3- Blood spread theory.***

***4- Lymphatic spread theory***

***5- Genetic theory.***

- Risk of endometriosis is seven times greater of a first degree relative has been affected by endometriosis.

-A relative risk of 75% has been noted in monozygotic twins of pt. with endometriosis (concordance).

- There is (+ve) relationship between endometriosis & other autoimmune disease e.g. SLE.

- Oxegene study (Oxford endometriosis - gene study) demonstrates:

(I) Polymorphism with GALT – enzyme galactose -1- phosphate uridly transferase gene locus.

(II) Polymorphism with glutathione S-transferase MI (GSTMI) gene, which is essential for detoxication of dioxin which become a risk factor for endometriosis.

***6- Cell - Mediated immunity Theory:***

Its reported alterations in cell – mediated immunity resulting in macrophage activation as one of the mechanisms possibly involved in the pathogenesis of endometriosis. Alterations in substances secreted by activated macrophages, including prostaglandins, interleukin I, and tumor necrosis factor, have also been found in the peritoneal fluids of women with endometriosis. These findings suggest that soluble products of activated macrophages secreted in peritoneal fluid may play a role in the decreased fertility associated with early stages of endometriosis. This effect might occur through alterations in ovum capture, sperm function, sperm oocyte interaction, or embryo cleavage rate.

**Predisposing factors.:**

1- Hyperestrogenism state: mention causes → Physiological / Pathological / Pharmacological

2-Cervical obstruction e.g. with cervical stenosis→predispose to retrograde menstruation.

3- Operation of endometruim in premenstrual or during menstrual phase e.g. D and C, HSG.

4- Females with high socioeconomic state

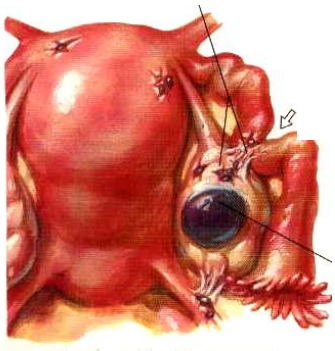
5- Some authors consider familial tendency as predisposing factor for endometriosis.

**Pathology :** → Ovarian endometriosis / Pelvic endometriosis.

**A) <u>Endometriosis of ovaries</u>:**

**I- Mac:**

- Small implants of endometriosis (dark spots = powder burns)
- Large cysts containing blood (chocolate cyst)



**Chocolate cyst**

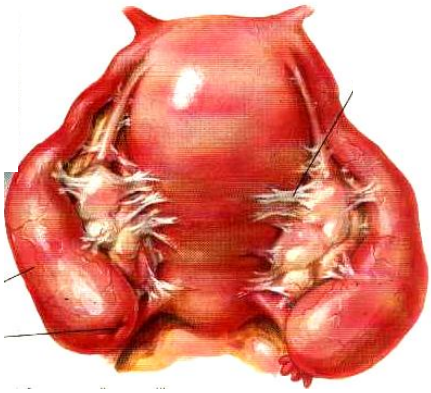
**II- Mic:** endometrial tissues (gland + stroma)

**B) <u>Pelvic endometriosis</u> : I-Mac:**

1- Multiple small dark spots especially in Douglas pouch which surrounded by adhesions due to:

a- peritoneal irritation by monthly bleeding.

b- associated chlamydial infection.



2- Extensive adhesion may lead to :

- Fixed RVF uterus.
- Frozen pelvis (especially on left side, why?)

**II- Mic:** endometrial tissues (gland + stroma).

**Complications of pelvic endometriosis:**

1- Infertility, ectopic pregnancy or recurrent abortions.

2- Extension to bladder → bleeding inside bladder at time of menstruation → menuria.
3- Extension to ureter → ureteric obstruction.
4- Extension to intestine → intestinal obstruction, to rectum → bleeding per rectum with time of menses.
5- Malignant changes; uncommon and usually occurs in endometrioma of ovary → can cause carcinoma or sarcoma.

**C.P:**

**I- Age and parity:** - Age: 30- years - Parity: nullipara
- Main C/O: infertility and pelvic pain.

**II- Symptoms:**

**[A] No symptoms:** and disease can be discovered accidentally on laparotomy or laparoscopy.

**[B] Symptomatic →** pelvic congestive syndrome.

1) *Pain*: the single main symptom on the form of :

1- Dysmenorrhea: - Starts few days before period of menses, and ↑ during period due to distention of ectopic glands with blood.
- Grinding in character. - Away from midline.

2- Deep dysparunia. 3-Deep seated pelvic pain.
4- Dysuria. 5- Dyscazia.
6- Low backache. 7- Sciatica.
8- Renal colic. 9- Loin pain.
10- Acute abdominal pain due to sudden rupture of endometrioma.

2) *Bleeding*: on the form of:

a- Menorrhagia: called persistent menorrhagia.
b- Polymenorrhea.

***Causes of menorrhagia or polymenorrhea with endometriosis:**

a- Hormonal imbalance.
b- Pelvic congestion and vascularity.
c- ↑ surface area of endometrium.

d- Associated endometrial hyperplasia.

3) *Infertility*: 30% of infertile female has endometriosis.

*** Causes of infertility in endometriosis:**

1- Ovary: - unovulation.
2- Trapping of ovum in periovarian adhesions.
3- ↑ $PGF_2 \propto$ in peritoneal fluid leads to
   a- unovulation. b- LPD
   c-↑ tubal motility. d- ↑ uterine motility.
4- ↑ Peritoneal macrophage → phagocyte sperms and ovum.
5- Loss of ovum pick up mechanism.
6- Disturbed tubal motility.
7- Chronic salpingitis (30% of cases).
8- Bilateral cornual block with endometrial implants.
9- Prevent implantation.
10- Fixed RVF uterus.
11- Deep dysparunia.
12- Associated hormonal imbalance.

4) *Increase vaginal discharge (leucorrhea).*

**[C] Symptoms of complications:** e.g → Menuria
→ (Bleeding per rectum)

**III-Signs:**

**[A] General examination:** patient usually is obese (BMI, waist to hip ratio).

**[B] Local examination:**

1- Ovarian endometriosis: may be unilateral or bilateral ovarian cyst.
2- Multiple nodules in Douglas pouch and when uterosacral ligament are involved → thickened = Banjo strings (cobble stone appearance).
3- Uterus firm in consistency may be fixed R.V.F.

**Investigation:**

1- U/S, CT scan or MRI on pelvis to show larger endometrioma.

2- Laparoscopy: remains 1st tool for diagnosis, staging, treatment even follow – up for endometriosis.

According to laparoscopy we have different methods for classification of endometriosis.

a- (American fertility society) staging (AFS) based on site of endometriosis lesion and adhesion, patient is given score for every site and total score is calculated (max. score is 54).

b- Acosta-staging:

- Mild endometriosis (stage I)
  1- Small scattered nodules in Douglas pouch.
  2- No scarring.
  3- No adhesions.
  4- No ovarian involvement.
- Moderate endometriosis (stage II):
  1- ↑↑ nodules in Douglas pouch.
  2- Scarring and puckering.
  3- +ve adhesion especially in Douglas pouch.
  4- Ovarian endometrioma < 2.5 cm in diameter.
- Severe endometriosis (stage III):
  1- Obliterated Douglas pouch.
  2-Severe scarring.
  3- Severe adhesion.
  4- Ovarian Endometrioma > 2.5 cm in diameter.
- Extensive endometriosis (stage IV):
  As (III) + involvement of bladder or rectum

3- Study of peritoneal fluid by laparoscopy:

a) ↑ peritoneal fluid leptin.

b) ↑ peritoneal fluid interleukin.

c) ↑ peritoneal fluid prostaglandin.

d) ↑ peritoneal fluid TNF.

e) ↓ peritoneal fluid platelet activating factor acetylhydrolase which leads to ↑ half life of platelet activating factor → play a key role in infertility associated with mild endometriosis.

4-Androgen therapeutic test: administration of androgen → dramatic relief of symptoms and signs → diagnosis of endometriosis

5- CA-125: tumour marker which ↑ in endometriosis and can be used for monitoring success of treatment and denote recurrent endometriosis. But it's non-specific as it ↑↑ with many diseases especially ovarian carcinoma.

6- Investigation for complications:

a- Cystoscopy b- IVP

c- Sigmoidoscopy d- Barium enema.

**Treatment of endometriosis:**

**I- Prophylactic ttt:**

1- Avoid any pelvic manipulation during menstruation .
2- Early treatment of cryptomenorrhea.
3- Treatment of hyperestrogenic state.

**II- Active treatment:**

***a-Expectant treatment:*** In young patient (wants pregnancy) with minimal *C/P.* and minimal lesion.

1- Encourage patient to get pregnancy .
(pregnancy is the optimum prophylaxis and curative for endometriosis).
2- Symptomatic treatment → anti PG.
3- Follow-up /6 months.

**b- *Hormonal treatment*:** → main line of treatment

- Indicated for marked symptoms with minimal lesion.
- Can be used preoperative to facilitate dissection and decrease incidence of recurrence.
- Can be used post operative for persistent or recurrent lesion.

All drugs used are antiestrogenic:

**1-Progesterone:**

-Mechanism: It induces pseudo-pregnancy state → atrophy of endometrial gland by:

1- Progesterone has anti E effect by:

a- It ↓ E- receptors in endometrial cells.

b- ↑ E metabolism inside liver.

2- Progesterone acts on endometriosis → decidual reaction → on continous use → exhaustion atrophy of glands.

-Dose:10-40 mg daily (progesterone) for 6-24 month.

**2- Dydrogesterore (Duphastone)** 10 mg tab 2 x 1 for 6-9 months.

- Duphaston is structurally & pharmacologically almost identical to natural endogenous progesterone with no androgenic & osetrogenic properties & no sedation & drowziness of other oral progesterone (uterogestan).

**3- Oral contraceptive pills:**

-Mechanism: as progesterone → pseudopregnancy → atrophy of endometriosis implants.

-Dose: use progesterone dominant preparation e.g. (ovral).

**4- Androgens** (obsolete):

Has very rapid and dramatic effect but with many side effects.

**5- Danazol:**

Chemistry:

- It's the most common drug used now for treatment of endometriosis.
- It's isoxazole derivative of ethinyl testosterone.

Mechanism: - Danazol increases androgenic hormonal environment by:

1- It's androgenic in nature.

2- By action on liver, it decrease sex hormone binding globulin → increase free circulating androgens.

- Danazol decreases estrogenic hormonal environment by.

1- It inhibits release of GnH from anterior pituitary.

2- It inhibits ovarian steroidogenesis →↓ E synthesis.

Dose: 200-800 mg/d for 6-9 months.

Side effects: 1-Pseudomenopausal action → Amenorrhea + menopausal symptoms (hot flush, dry vagina).

2- Virilizing effects of androgens: weight gain, acne, ……

**6- Gestrinone:** The same mechanisms of danazol with advantages:

(1) Long half life (28 hr after oral intake). So can be given twice / week.

(2) As potent as danazol with less side effect especially menopausal syndrome.

**7- 5-Gn-RH- analogue:**

- In pharmacological dose → It inhibit axis e.g. Tryptorelin (decapeptyl) injection I.M.I. / month.

- The most serious side effect is osteoporosis which occur when serum E2 level < 20 pg/ml for longtime, which occurs with GnRH against treatment. To counteract this effect we can use (add – back) / strategy with GnRh–a in treatment of endometriosis to ↓ demineralization with no objective effect on efficacy of symptom control of endometriosis e.g. (tibolone, bisphosphonate, calcitonin, P.T.H, SERM.)

***N.B.1:*** Guidelines for use of GnRh-a or danazol in treatment of endometriosis:

1- Document +ve of endometriosis by laparoscopy.

2- Ensure that pt. is not pregnant.

3- Employ the drug at a dose which cause amenorrhea.

4- Treatment for at least 6 months.

5- Monitor side effects carefully especially:

- Danazol: For side effects of increased androgen.
- GnRh-a: For osteoprosis.

**N.B.2:** Comparison between danazole and GnRh-a in the treatment of endometriosis:

1- Both can cause hypoestrogenism but it is more and consistent with GnRh-a than danazol. Usually with GnRh-a the level of $E_2$ may reach < 20 pg/ml like postmenopausal level (medical castration).while with danazole the level is 30 – 45 pg/ml.

2- Danazol only of androgenic activity → side effects (virilization, acne, weight gain, decrease HDL with increase risk of cardiovascular disease, break through bleeding.

But on other side androgen may decrease risk of osteoporosis.

3- Both cause symptomatic relief but GnRh-a is more effective esp. in severe cases of endometriosis.

4- Danazol used orally, while GnRh-a never used orally:

**N.B.3:** Efficacy of medical treatment of endometriosis:

1- Pain : all drugs are very effective in control of pain either progestins, danazol, gestrinone, GnRh-a.

2- Conception is either impossible or contraindicated during medical treatment of endometriosis.

Again some describe that : No evidence that medical treatment of minimal to mild endometriosis leads to better chances of pregnancy than expectant treatment.

**C- *Surgical treatment*:**

- Indicated for marked symptoms with marked lesion:

### ➢ Conservation surgery

Aim**:** preserve uterus + one ovary +one tube

Age**:** young patient + want pregnancy usually with ovarian endometriosis

Technique**:** mainly laparoscopy:

1- Ovarian cyst → ovarian cystectomy.
2- Pelvic adhesions → adhesolysis.
3- Tubal block → salpingostomy.

4- Any pelvic or ovarian nodules: → fulgration by diathermy or laser.

***D- Gene therapy:*** against - GALT - GST-MI.

## Adenomyosis

**Definition :** Presence of endometrial tissues between uterine muscles.

**Theory :** **Cullen's diverticular theory:**

Some of endometrial glands grow deeply between muscle fibres of uterus. This explain adenomyosis .

**P.F :** Any endometrial scar as C.S or previous endometritis.

**Pathology :**

**I-Mic:** Presence of endometrial tissues (gland + stroma) in between myometrial tissues.

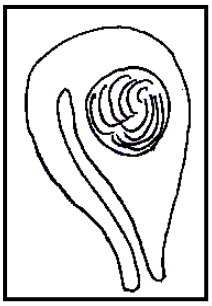
**II- Mac:**

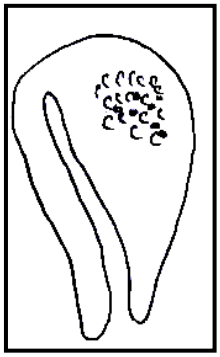
1- Uterus is firm in consistency.

2- Increased length of uterus detected by uterine sound.

3- Usually symmetrical enlargement of uterus, but not more than 12 week. resulting from diffuse thickening of myometruim due to irritation by ectopic endometruim.

Occasionally a localized area of adenomyosis may be present which must be differentiated from fibroid by:





| | |
|---|---|
| - Capsule in fibroid | - No capsule in localized adenomyosis |
| - Whorly appearance | - Coarse reticular pattern |
| - Pale in colour | - Darker in colours |

**C.P :**

1- Adenomyosis age: 40 year parity: multipara.
2- Main C/O: abnormal uterine bleeding, chronic pelvic pain.
3- Adenomyosis: usually symmetrical enlarged uterus (Bulky uterus = up to 10-12 week size).

**Investigation:** U/S, CT scan or MRI on uterus.

**Treatment:** As endometrial tissues → of basalis type, so not respond to hormones, main line of ttt of adenomyosis is surgical ttt.

**1- Hormonal treatment:** of little value

**2- Radical surgery:** TAH + BSO to achieve complete cure.

**N. B.: Sum**. Difference between adenomyosis and pelvic endometriosis

| Adenomyosis | Endometriosis |
|---|---|
| **Definition:** endometrial glands between myometruim. | - endometrial glands outside uterus |
| **Thories:** Cullen's diverticulem. | - Metaplasia or retrograde menses. |
| **Pathology:** basal glands which do not respond to hormones → small cysts. | - Superficial parts of gland respond to hormone → larger cysts. |
| **Age:** old age, multipara | - young age, nullipara. |
| **ttt: mainly hysterectomy as:**<br>a- No response to hormones<br>b- old age and multipara | - mainly hormonal |